

All random variables appearing below are assumed to be defined with respect to a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Let $Z_1, Z_2, \dots \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$. Consider a TPM P on a finite set $\mathcal{X}$, and let $f : \mathcal{X} \times (0, 1) \rightarrow \mathbb{R}$ be the random representation map associated with P . Let $X_1 \sim \mu$ be independent of Z_1 , where μ is a fixed PMF on $\mathcal{X}$, and let

$$X_n = f(X_{n-1}, Z_n), \quad n \geq 2.$$

Prove that $\mathbf{X} = \mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a homogeneous DTMC with TPM P .

- Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a homogeneous DTMC on a finite state space $\mathcal{X}$ with TPM P . Let $\{Y_n\}_{n \in \mathbb{N}}$ be another process defined as

$$Y_n = X_{Kn}, \quad n \in \mathbb{N},$$

where $K \in \mathbb{N}$ is a fixed constant.

Prove that $\{Y_n\}_{n \in \mathbb{N}}$ is a homogeneous DTMC and identify its TPM.

- Suppose that the numbers of families that check in to a hotel on successive days are independent Poisson random variables with a fixed parameter $\lambda > 0$. Also, suppose that the number of days that any given family stays in the hotel is a Geometric random variable with parameter $p \in (0, 1)$. (Thus, a family who spent the previous night in the hotel will, independently of how long they have already spent in the hotel, check out the next day with probability p .) Assume that all families act independently of one another.

- If X_n denotes the number of families present in the hotel at the end of day n , show that $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a time-homogeneous DTMC. You may assume that $X_1 = 0$.

Hint: Random mapping representation.

- Identify the TPM of the process $\mathbf{X} = \mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$.

- (Eigenvalues and eigenvectors of a row stochastic matrix)**

Let P be a row-stochastic matrix of size $d \times d$ for some fixed $d \in \mathbb{N}$. For any $i \in \{1, \dots, d\}$, let

$$R_i := \sum_{j \neq i} P(i, j).$$

- Let λ be an eigenvalue of P with an associated eigenvector $\mathbf{x} = [x_1, \dots, x_d]^\top$. Let

$$i^* := \arg \max_i |x_i|$$

denote the coordinate of $\mathbf{x}$ with the largest absolute value. If i^* is not unique, simply pick i^* at random from the set of all indices which attain the maximum in the expression above. Show that

$$|\lambda - P(i^*, i^*)| \leq R_{i^*}.$$

Hint: Consider the equation

$$(P - \lambda I) \mathbf{x} = \mathbf{0}.$$

Rearrange terms in the above equation by having i^* term on the left-hand side and remaining $i \neq i^*$ terms on the right-hand side, apply triangle inequality, and use the fact that $|x_i|/|x_{i^*}| \leq 1$ for all $i \neq i^*$.

- Conclude from part (a) that if λ is any eigenvalue of P , then $|\lambda| \leq 1$.
- If P has strictly positive entries ($P(i, j) > 0$ for all $i, j \in \mathcal{X}$), then show that $\lambda = 1$ is a simple eigenvalue (i.e., its eigenspace has dimension 1).

Hint:

We know that $\mathbf{1} = [1, \dots, 1]^\top$ is an eigenvector corresponding to eigenvalue $\lambda = 1$. We need to demonstrate that any other eigenvector corresponding to $\lambda = 1$ is only a scaled version of $\mathbf{1}$, of the form $c\mathbf{1}$. Assume that there exists an eigenvector $\mathbf{x} = [x_1, \dots, x_d]^\top$ in which $x_i \neq x_j$ for some $i \neq j$, and try come up with a contradiction.

- (d) Continuing on part (c), assuming that P has all entries strictly positive, show that if $\lambda' \neq 1$ is any other eigenvalue of P , then $|\lambda'| < 1$.

5. Let $\{X_n\}_{n \in \mathbb{N}}$ be a time-homogeneous DTMC on the state space $\mathcal{X} = \{0, 1, 2, 3, 4, 5\}$, with the TPM as shown below.

$$\begin{array}{c} \begin{matrix} & 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{8} & 0 & \frac{7}{8} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{3}{4} & 0 & \frac{1}{4} & 0 \\ 0 & \frac{1}{5} & 0 & \frac{1}{5} & \frac{1}{5} & \frac{2}{5} \end{pmatrix} \end{array}$$

- (a) Classify the states as transient and recurrent.
(b) Let $T_{\{0,1\}}$ be defined as the random variable

$$T_{\{0,1\}} := \inf\{n \geq 2 : X_n \in \{0, 1\}\}.$$

For each $x \in \{0, 1, 2, 3, 4, 5\}$, determine $\mathbb{P}(\{T_{\{0,1\}} < +\infty\} \mid \{X_1 = x\})$.

6. There are N empty boxes and a countable infinite collection of balls.
Let $X_1 = N$. At each time step $n \geq 2$, a box is chosen uniformly at random and a ball is placed in the chosen box.
Let X_n denote the number of empty boxes at the end of time step n .
(a) Show that $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a DTMC.
(b) Identify the state space and specify the TPM of $\mathbf{X}$.
(c) Classify the states as transient, null recurrent, and positive recurrent.
7. Fix $N \in \mathbb{N}$, and let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a homogeneous DTMC on the state space $\mathcal{X} = \{0, 1, \dots, N\}$, with transition probabilities given by

$$P(x, y) = \begin{cases} 1 - \frac{x}{N}, & y = x + 1, \\ \frac{x}{N}, & y = x - 1. \end{cases}$$

Show that the above TPM is irreducible and positive recurrent, and compute μ_{xx} for all $x \in \mathcal{X}$.

8. **Extension of Question 6, Homework 02:**

Let $\{\xi_n\}_{n \in \mathbb{N}}$ be a sequence of i.i.d. random variables with

$$\mathbb{P}(\{\xi_n = +1\}) = p = 1 - \mathbb{P}(\{\xi_n = -1\}),$$

for some $p \in (0, 1)$. Let $S_0 = 0$, and for each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^n \xi_i$ denote the partial sum at time n .

- (a) Show that $\mathbf{S} = \{S_n\}_{n \geq 0}$ is an irreducible and homogeneous DTMC on $\mathcal{X} = \mathbb{Z}$.
(b) If $p > \frac{1}{2}$, show that

$$\mathbb{P}\left(\left\{\liminf_{n \rightarrow \infty} S_n = +\infty\right\}\right) = 1.$$

Hence conclude that all states in $\mathcal{X}$ are transient.

- (c) If $p < \frac{1}{2}$, show that

$$\mathbb{P}\left(\left\{\limsup_{n \rightarrow \infty} S_n = -\infty\right\}\right) = 1.$$

Hence conclude that all states in $\mathcal{X}$ are transient.

(d) In either of the cases (b), (c), show that no invariant distribution exists for the TPM of $\mathbf{S}$.

(e) Suppose that $p = \frac{1}{2}$. Show that

$$\mathbb{P}\left(\left\{\liminf_{n \rightarrow \infty} S_n = -\infty\right\}\right) = 1, \quad \mathbb{P}\left(\left\{\limsup_{n \rightarrow \infty} S_n = +\infty\right\}\right) = 1,$$

and hence conclude that every state in $\mathcal{X}$ is recurrent.

(f) Continuing on part (e), show that no invariant distribution exists for the TPM of $\mathbf{S}$ when $p = \frac{1}{2}$, and thereby conclude that all states must be null recurrent.